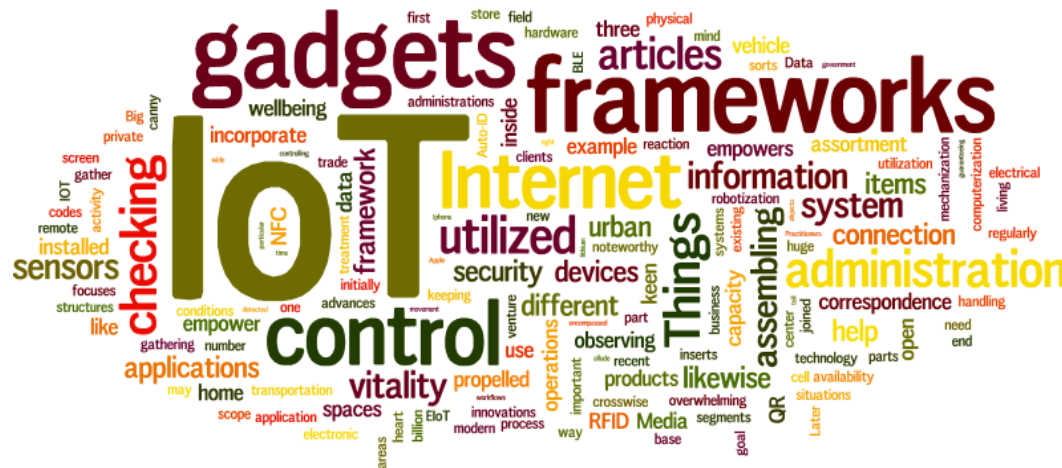


# CS578: Internet of Things

## The “Things” in IoT



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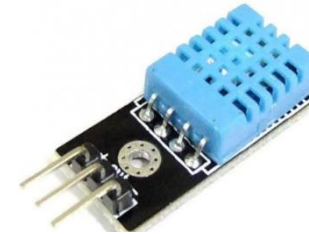
E-mail: [manaskhatua@iitg.ac.in](mailto:manaskhatua@iitg.ac.in)

*“Arise! Awake! Stop not till the goal is reached.”- Swami Vivekananda.*

# The “Things”

➤ **Sensors & Actuators** are the fundamental building blocks of IoT

- Sensor senses
- Actuator acts



Sensor

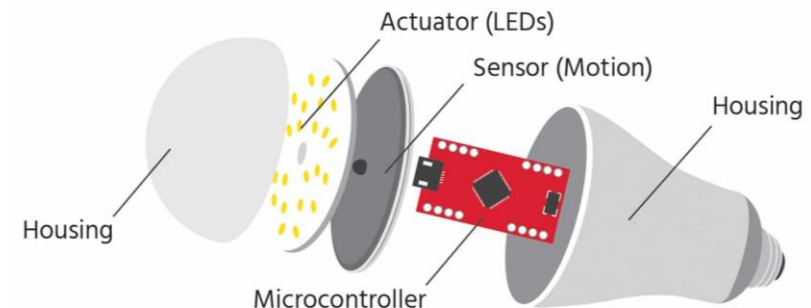
➤ **Smart objects** are any physical objects that contain

- Embedded technology
  - Microcontroller unit, memory storage, power supply, communication ports, input and output, timer or counter
- Sensors and/or actuators



Actuator

- Smart objects are to sense and/or interact with their environment in a meaningful way
  - being interconnected, and
  - enabling communication among themselves or with external agent.



Smart Object

# Sensors

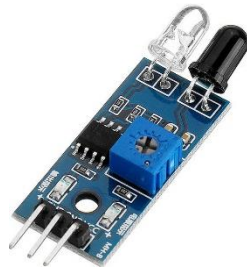
- It **measures some physical quantity** and converts that measurement into analog/digital form
- There are a number of ways to group and cluster sensors into different categories
  - Based on **external energy requirement**
    - Active / Passive
  - Based on **placement location**
    - Invasive / Non-invasive
  - Based on **distance from the sensing object**
    - Contact / No-contact
  - Based on **sensing mechanism**
    - Thermoelectric / Electromechanical / Piezo resistive / Optic / Electric / Fluid mechanics / Photoelastic / etc.
  - Based on **sensing parameter**
    - Position / Occupancy / Motion / Velocity / Force / Pressure / Flow / Humidity / Light / Temperature / Acoustic / Radiation / Chemical / Biosensors / etc.
  - Based on **application industry**
    - Medical / Manufacturing / Agriculture / etc.
  - Based on **measuring scale**
    - Absolute / Relative

# Sensor Types: What it measures

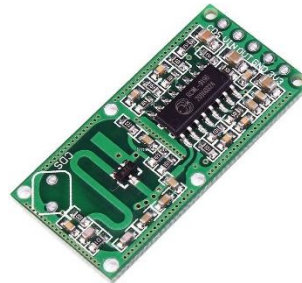
| Sensor Type      | Description                                                                                                                                                                                    | Example                                                                                                     |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <u>Position</u>  | <ul style="list-style-type: none"><li>Measures the position of an object</li><li>Position could be absolute/relative</li><li>Position sensor could be linear, angular, or multi-axis</li></ul> | <ul style="list-style-type: none"><li>Proximity sensor</li><li>Potentiometer</li><li>Inclinometer</li></ul> |
| <u>Occupancy</u> | <ul style="list-style-type: none"><li>Detects the presence of people and animals in a surveillance area</li><li>Generates signal even when a person is stationary</li></ul>                    | <ul style="list-style-type: none"><li>Radar Sensor</li></ul>                                                |
| <u>Motion</u>    | <ul style="list-style-type: none"><li>Detects the movement of people and objects</li></ul>                                                                                                     | <ul style="list-style-type: none"><li>Passive Infrared (PIR) Sensor</li></ul>                               |



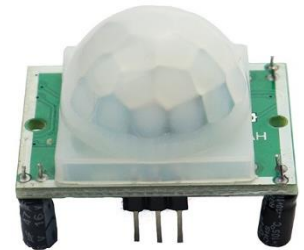
Ultrasonic Proximity Sensor



Infrared Proximity Sensor



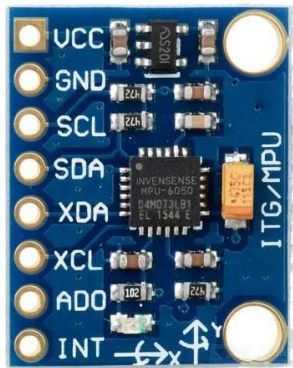
Microwave Radar Sensor



PIR Motion Sensor

# Cont...

| Sensor Type                 | Description                                                                                                                                                       | Example                                                                              |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Velocity and Acceleration ✓ | <ul style="list-style-type: none"> <li>Velocity sensor measures how fast an object moves</li> <li>Acceleration sensor measures the changes in velocity</li> </ul> | <ul style="list-style-type: none"> <li>Gyroscope</li> <li>Accelerometer</li> </ul>   |
| Force ✓                     | <ul style="list-style-type: none"> <li>Detects whether a physical force is applied and the magnitude of the force</li> </ul>                                      | <ul style="list-style-type: none"> <li>Tactile sensor</li> <li>Viscometer</li> </ul> |
| Pressure ✓                  | <ul style="list-style-type: none"> <li>Measuring the force applied by liquids or gases</li> <li>It is measured as force per unit area</li> </ul>                  | <ul style="list-style-type: none"> <li>Barometer</li> <li>Piezometer</li> </ul>      |



Gyroscope



Capacitive Touch Sensor



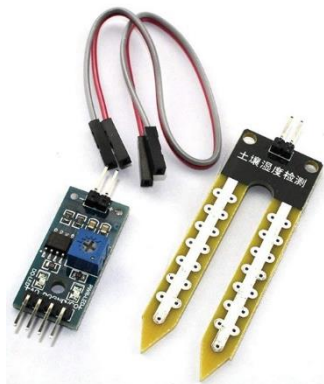
Barometric Pressure Sensor

# Cont...

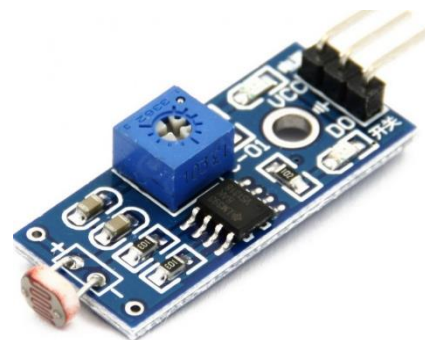
| Sensor Type | Description                                                                                                                                      | Example                                                                                                           |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Flow        | <ul style="list-style-type: none"><li>• Detects the rate of fluid flow through a system in given period of time</li></ul>                        | <ul style="list-style-type: none"><li>• Water meter</li><li>• Anemometer</li></ul>                                |
| Humidity    | <ul style="list-style-type: none"><li>• Detects amount of water vapour in the air</li><li>• Can be measured in absolute/relative scale</li></ul> | <ul style="list-style-type: none"><li>• Hygrometer</li><li>• Soil moisture sensor</li></ul>                       |
| Light       | <ul style="list-style-type: none"><li>• Detects the presence of light</li></ul>                                                                  | <ul style="list-style-type: none"><li>• LDR light sensor</li><li>• Photodetector</li><li>• Flame Sensor</li></ul> |



Water meter



Soil moisture sensor



LDR light sensor



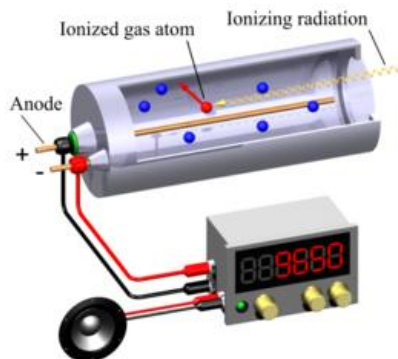
Flame sensor

# Cont...

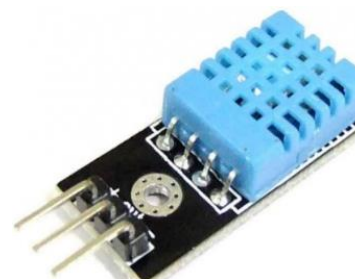
| Sensor Type | Description                                                                                                                                           | Example                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Radiation   | <ul style="list-style-type: none"><li>• Detects the radiation in the environment</li></ul>                                                            | <ul style="list-style-type: none"><li>• Neutron detector</li><li>• Geiger-Muller counter</li></ul>              |
| Temperature | <ul style="list-style-type: none"><li>• Measures the amount of heat or cold present in the system</li><li>• Two type: contact / non-contact</li></ul> | <ul style="list-style-type: none"><li>• Thermometer</li><li>• Temperature gauge</li><li>• Calorimeter</li></ul> |



Neutron detector



Geiger-Muller counter



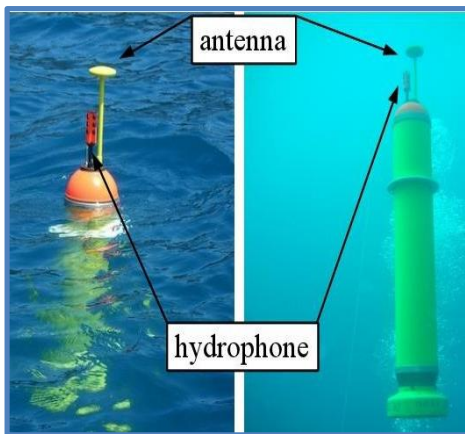
Temperature  
Sensor



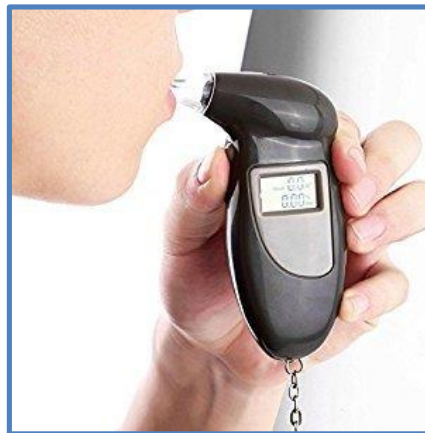
Thermo-  
Hygrometer

# Cont...

| Sensor Type | Description                                                                                                                                                       | Example                                                                                                                             |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Acoustic    | <ul style="list-style-type: none"> <li>Measures sound level</li> </ul>                                                                                            | <ul style="list-style-type: none"> <li>Microphone</li> <li>Hydrophone</li> </ul>                                                    |
| Chemical    | <ul style="list-style-type: none"> <li>Measures the concentration of a chemical (e.g. CO<sub>2</sub>) in a system</li> </ul>                                      | <ul style="list-style-type: none"> <li>Smoke detector</li> <li>Breathalyzer</li> </ul>                                              |
| Biosensor   | <ul style="list-style-type: none"> <li>Detects various biological elements, such as organisms, tissues, cells, enzymes, antibodies, nucleic acid, etc.</li> </ul> | <ul style="list-style-type: none"> <li>Pulse oximeter</li> <li>Electrocardiograph (ECG)</li> <li>Blood glucose biosensor</li> </ul> |



Hydrophone

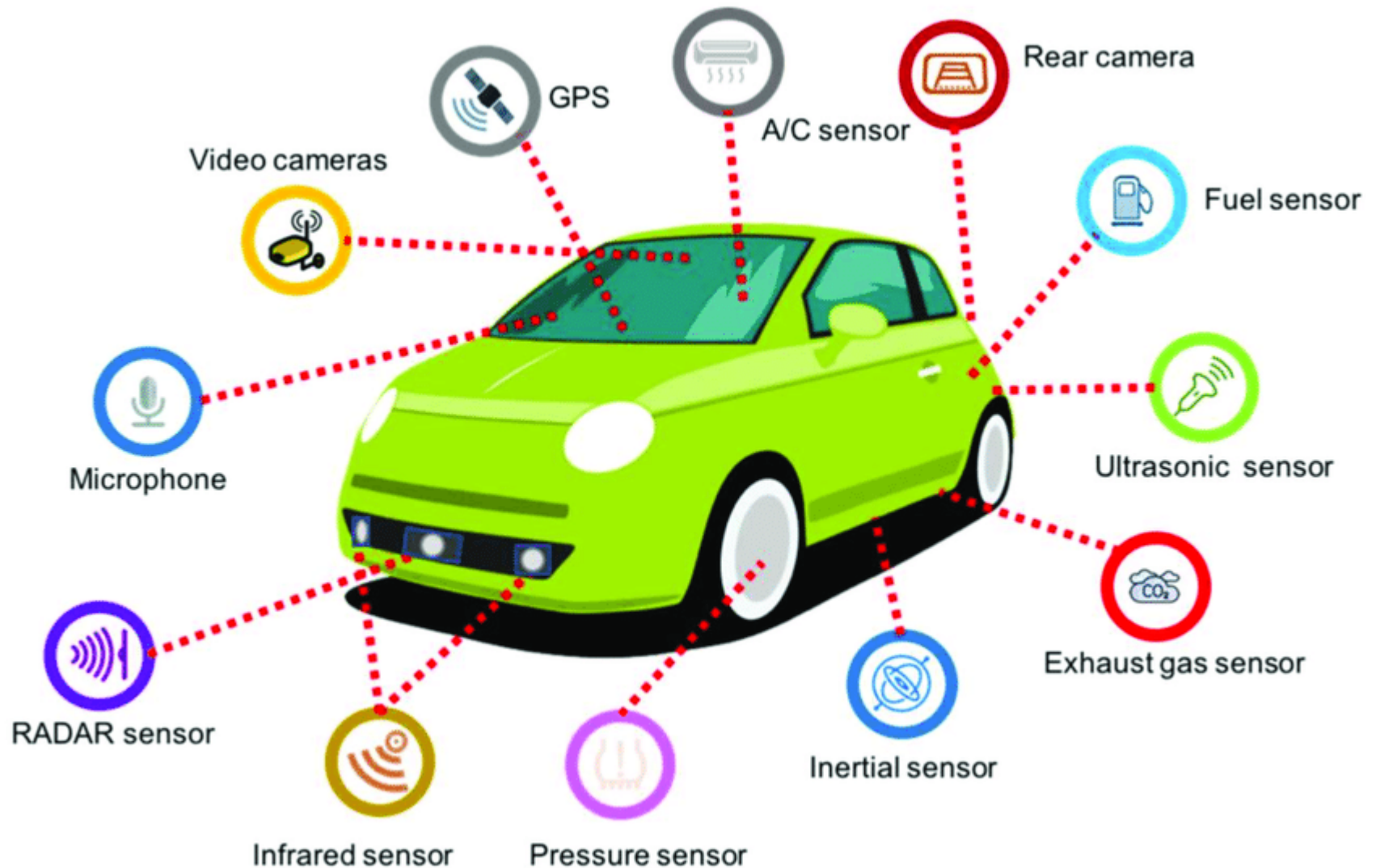


Breathalyzer



Pulse oximeter

# Sensors in a Smart Car

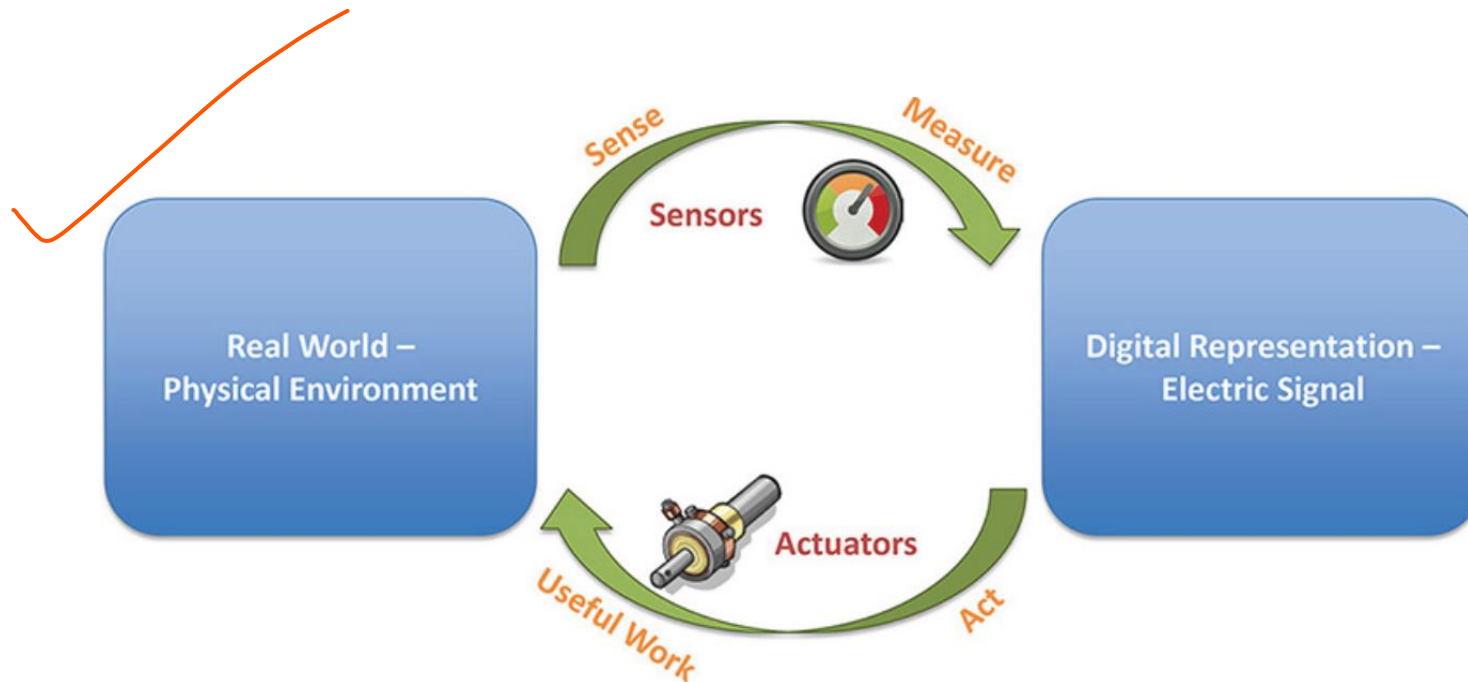


# Sensors in a Smartphone



# Actuators

- Sensors are designed to sense and measure the surrounding environment
- Actuators receive some type of control signal (commonly an electrical signal or digital command) that **triggers a physical effect**, usually some type of motion, force, and so on.



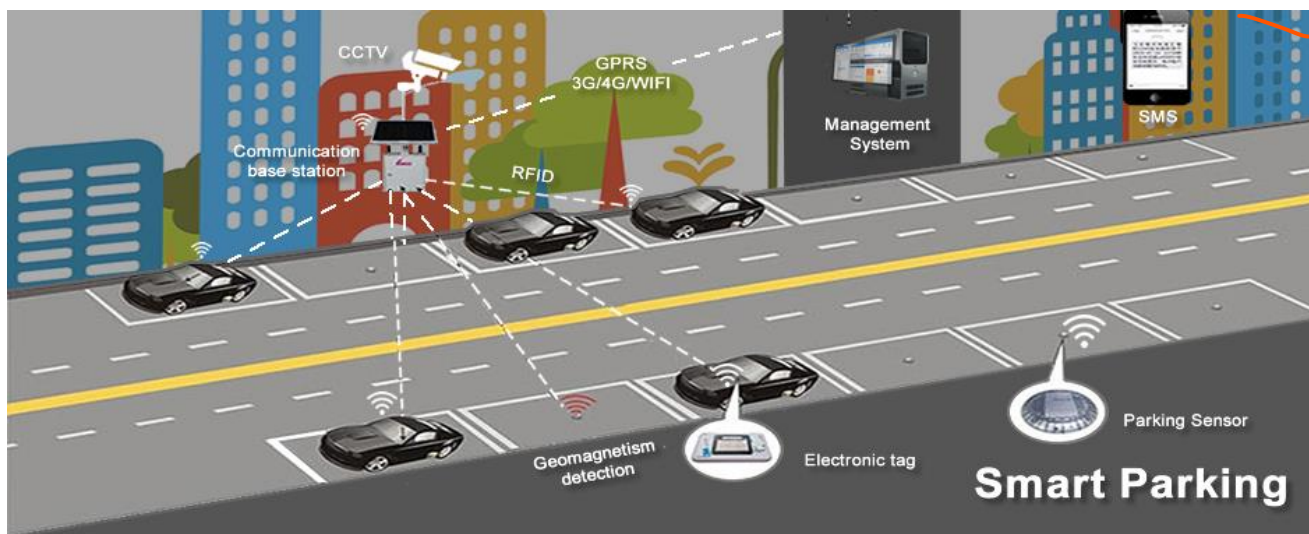
Source: [https://cdn2.hubspot.net/hubfs/1878050/Landingpages/Events/Schwabengipfel/Guido\\_Schmutz\\_IoT-Cloud-or-OnPrem.pdf?t=1501051153000](https://cdn2.hubspot.net/hubfs/1878050/Landingpages/Events/Schwabengipfel/Guido_Schmutz_IoT-Cloud-or-OnPrem.pdf?t=1501051153000)

# IoT based Automated Systems

## Smart Parking System

Source:

<https://www.mobiloitte.com/blog/smart-parking-solution-using-iot/>



## Smart Irrigation System

Source:

<https://www.hydropoint.com/what-is-smart-irrigation/>



# Actuator Classification

- Common ways to classify actuators:
  - ✓ Type of motion they produce
    - e.g. linear, rotary, one/two/three axes
  - ✓ Power output
    - e.g. high power, low power, micro power
  - ✓ Binary / Continuous output
    - Based on number of stable-state outputs
  - ✓ Area of application
    - Specific industry or vertical where they are used
  - ✓ Type of energy
    - e.g. mechanical energy, electrical energy, hydraulic energy, etc.

# Actuators by Energy Type

| Type                                                              | Examples                                                                          |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Mechanical actuators                                              | Lever, Screw jack, Hand crank                                                     |
| Electrical actuators                                              | Thyristor, Bipolar transistor, Diode                                              |
| Electromechanical actuators                                       | AC motor, DC motor, Step motor                                                    |
| Electromagnetic actuators                                         | Electromagnet, Linear solenoid                                                    |
| Hydraulic and Pneumatic actuators                                 | Hydraulic cylinder, Pneumatic cylinder, Piston, Pressure control valve, Air motor |
| Smart material actuator (includes thermal and magnetic actuators) | Magnetorestrictive material, Bimetallic strip, Piezoelectric bimorph              |

# Smart Objects

- It is the building blocks of IoT
- Smart object has the following **five characteristics**:

- **Sensor(s) and/or Actuator(s)**

- **Processing unit**

- For acquiring sensed data from sensors,
- processing and analysing sensing data,
- coordinating control signals to any actuators, and
- controlling many functions (e.g. communication unit, power unit).

- **Memory**

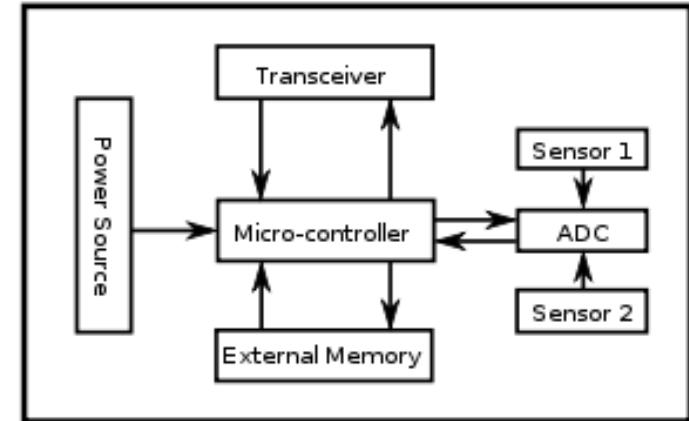
- Mostly on-chip flash memory
- user memory used for storing application related data
- program memory used for programming the device

- **Communication unit**

- Responsible for connecting a smart object with other smart objects and the outside world (via the network using wireless/wired communication)

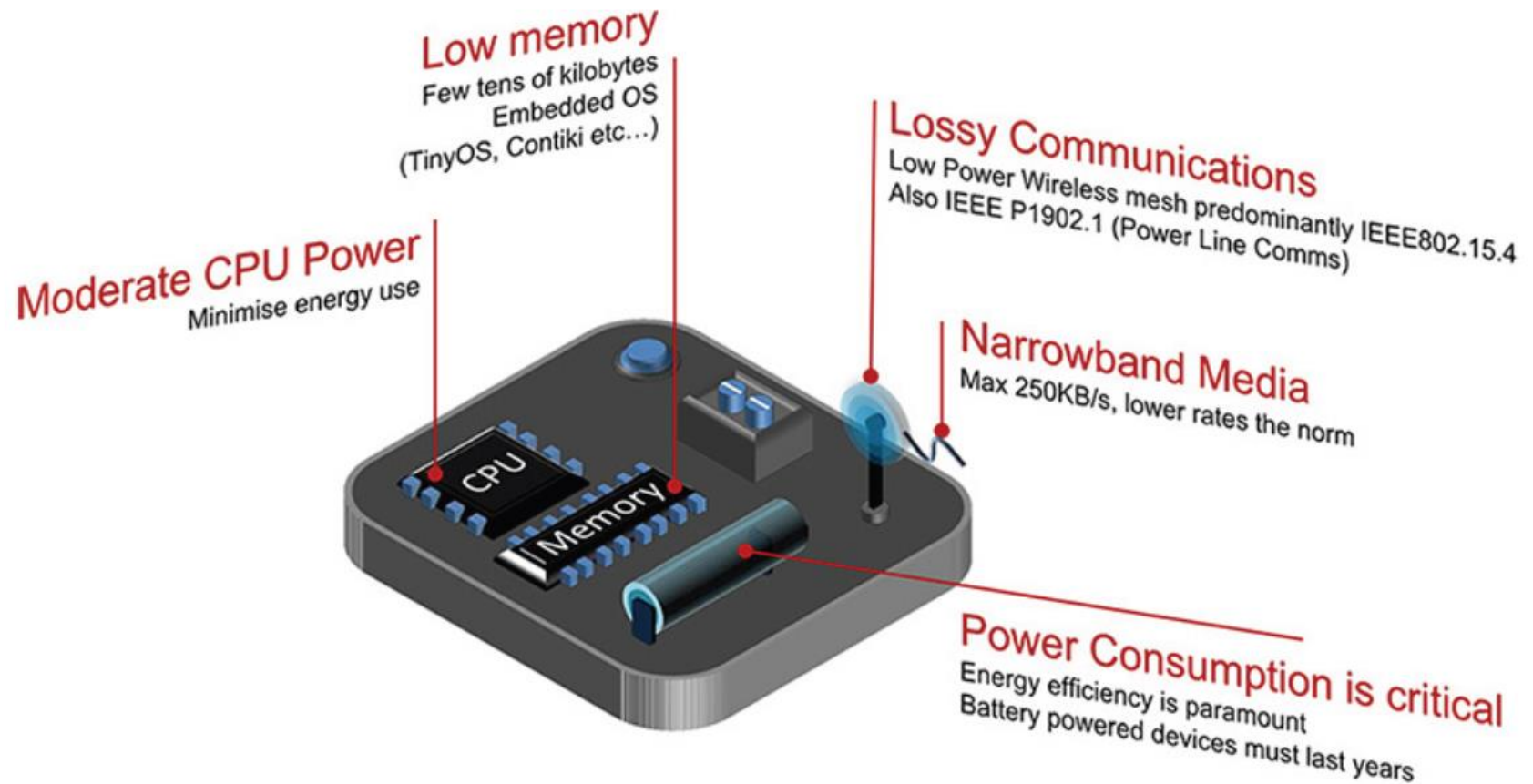
- **Power source**

- To powered all components of the smart object



TelosB Mote

# Cont...



Source: Cisco

# Present Trends in Smart Objects



- Size is decreasing
- Power consumption is decreasing
- Processing power is increasing
- Communication capabilities are improving
- Communication is being increasingly standardized

# Lessons Learned



- ✓ What is “Things” in IoT
- ✓ Classification method of Sensors
- ✓ Different Sensors based on Sensing parameter
- ✓ Classification method of Actuators
- ✓ Different Actuators based on Energy type
- ✓ What is “Smart object” in IoT

# Thanks!



Figures and slide materials are taken from the following Books:

1. David Hanes *et al.*, “**IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things**”, 1<sup>st</sup> Edition, 2018, Pearson India.